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Brief No. 73

Simulating the Future of European Security: Wargaming for Space Operations

Between 17 and 28 March 2025, France hosted the 5th edition of **AsterX**, which stands as **Europe's biggest wargaming activity involving space**. July 2024 also saw the UK conduct its first space wargame, "Space Warrior", featuring participants from the MoD, industry, and international allies.[1] The rapid digitisation and military operations have led to technological disruption becoming a preferred method of destabilisation, with the war in Ukraine exposing the truly multidimensional aspects of space operations. Pressure on policymakers to **reevaluate existing strategies and the continent's security parameters will only accelerate amid a tempestuous geopolitical environment**. Emerging threats and the risk of receding U.S. support to NATO have added further pressure to reevaluate existing strategies. Similarly, **Europe needs to better define a coordinated approach towards securing and protecting its space assets.** Wargaming, therefore, provides a key pillar for improving Europe's preparedness and better defining the processes and responsibilities in protecting space assets against potential threats.

> "The EU Space Threat Response Architecture (STRA) [...] should further evolve to allow the EU to answer to threats in space technically, politically, diplomatically or economically." (**Niinistö Report, 2024**)

1. Wargaming: A Tool for Resilience and Preparedness

The art of wargaming in its modern format is commonly argued to have emerged in the late 18th Century with the advent of *kriegsspiel*, invented by George Leopold von Reisswitz, which (in simplified terms), using a topographical map and physical pieces, allowed for visual testing and more realistic representation of strategic scenarios.[2]

Wargaming is often an umbrella term for various forms of conflict simulation. Wargaming acts as a practice ground to observe and test decision-making using pre-determined objectives (which can adapt during the game), creating a safe-to-fail environment to develop effective military tactics. The **purposes** of conducting wargames range from **understanding system capabilities and current tactics to educating military professionals and developing decision-making processes.**[3]



Figure 1: Primary differences between wargaming and live exercises

Wargaming allows participants to **become more aware of the inherent vulnerabilities of their tactics and concepts of operations** and, in turn, supports them in refining conceptual frameworks, such as the definition of red lines.[4] These aspects are **vital in understanding how to appropriately attribute, react, and respond to belligerent activity in space.** Unlike the other traditional warfighting domains, space suffers from a lack of experience and historical practice related to the management of threats.

The **limited clarification of a European doctrine in relation to responding to threats in space** means wargames can support establishing a definition for red lines and what constitutes aggressive

[1] Royal Air Force, "Space Command holds UK's first space wargame" 29 July 2024. (Link)
[2] Schuurman, Paul. 2017. "Models of War 1770–1830: The Birth of Wargames and the Trade-off between Realism and Simplicity." *History of European Ideas*, 43(5): 442–55. doi:10.1080/01916599.2017.1366928.
[3] This brief includes military exercises within its broader discussion on wargaming while acknowledging and aware of their distinction.
[4] Dudney, Robert S., 2011, "Hard Lessons at Schriever Wargame", *Air Force Magazine*. (Link).

behaviour. The EU Space Strategy for Security and Defence (EU SSSD), while noting a rise in threats in space, remains vague in its characterisation and response.

Moreover, the storytelling element is crucial for military space actors because, unlike military personnel from the land, sea, and air domains, who can physically reenact scenarios in the real world, **space personnel are not in situ actors**.[5] Besides ASATs and kinetic weapons, space assets are primarily subject to electronic and cyberattacks. This can result in a disconnect between the individual actor and the conflict itself. The experience gained through wargaming is particularly advantageous in comparison to live exercises by virtue of **minimised cost and reduced risk**.[6] Military actors would need to launch bespoke satellites into orbit for testing and training purposes in the format of live exercises. This is exemplified in the third edition of CYSAT where Thales managed to control the OPS-SAT through an ethical satellite hacking exercise.[7] **Training fleets for space assets do not exist to a similar extent as trainer aircraft fleets for air forces,** and wargames enable actors to test approaches without bearing the cost of launching training and demonstration fleets.

Another benefit of wargames is that they further support inclusivity amongst a broader set of stakeholders. As space continuously serves other military domains, it is important to be able to conduct **unhindered** coordination between terrestrial and space units. This can be facilitated in a wargaming setting, where actors from all domains can collaborate effectively under one roof and within the same environment.[8] Similarly, wargames present an **appropriate medium to include commercial space actors, which are becoming more prominent within national defence architectures** and activities.



Figure 2: Key advantages of wargaming for space actors

2. Putting theory into practice – examples of international wargames

The U.S. Shriever Wargame is one of the most notable wargaming activities that focuses on the use of space assets and operations. This has allowed the U.S. to better conceive the use of its space capacities through the inclusion of space operations in military wargames. While the U.S. has historically also led in developing capabilities to ensure on-orbit superiority, the current U.S. administration is signalling a **concerning yet decisive** turn toward treating space as an active warfighting domain, including through the **development of offensive capabilities**.

> "I feel like there's no way to ignore the fact that the next and the most important domain of warfare will be the space domain. … So, you're going to see far more investment from this administration into that domain, both offensively and defensively." **(Pete Hegseth, Secretary of Defence, U.S., 2025)**

As space assets become increasingly crucial in our everyday lives and security, it **becomes vital to ensure the resilience of Europe's space systems and actively protect them**. In the context of

[5] Perla, Peter P. and McGrady, ED. 2011. "Why Wargaming Works," *Naval War College Review,* 64(3) (Link)
[6] Perla, Peter P., and Branting, Darryl L. 1986. "Wargames, Exercises, and Analysis" *Center for Naval Analyses,* 86(20): i–12.
[7] Thales Group, "Thales seizes control of ESA demonstration satellite in first cybersecurity exercise of its kind", April 25, 2023, (Link)
[8] Dr. Rick Martin (host), Dr. Ellis Bartels, Brian Raymond. "Episode 102: Injecting Space into Wargaming", *The Space Policy Show*, The Aerospace Corporation. September 22, 2022 (Link).

the EU, an annual Space Threat Response Architecture Exercise (STRA-X) aims to test the EU's capacity to detect, attribute, and respond to attacks on its space assets.[9] STRA-X mobilises all relevant political, diplomatic and technical actors (such as the Galileo Security Monitoring Centre) and involves the EEAS, EU Member States, EUSPA, and DG DEFIS. The exercise is carried out at both the operational and political levels, using the lessons learned from the operational component (including incident detection and response) to reflect on outcomes and discuss with the Political and Security Committee (PSC) regarding the resilience of the EU's Space Threat Response Architecture.

NATO has approached using wargames as a tool for developing strategies. Researchers from the NATO Science and Technology Organisation and HQ Supreme Allied Command Transformation conducted the Multi-Domain Space Deterrence Framework 2024 wargame, held between 29 April and 2 May in the Netherlands, to develop and test a comprehensive space deterrence framework.[10]

The **most high-profile example of national wargaming and exercises in Europe is France's AsterX**, which has grown rapidly over its 4-year existence, recently including participants from 12 allied nations and growing from being a 5-day event in 2021 to the most recent edition lasting 12 days.[11] Moreover, in 2023, AsterX was integrated within the ORION 23 NATO exercise, supporting a multi-domain approach to military training and preparedness, and the wargame opted for a real-time approach, as opposed to time-lapse, to enhance tactical realism.[12] This growth conveys how the French Armed Forces are taking leadership in response to the need for more training and better coordination in space operations. The UK has also recognised the benefit of space in its Defence Space Strategy, calling for the inclusion of space among national wargames and the integration of space into military exercises alongside land, maritime and air domains.[13] "With the Space Warrior wargame marking a first step, the UK could increasingly leverage wargaming to deepen practical defence collaboration with other European states.

3. Wargaming in Europe: United Front or Fragmented Effort?



Figure 3: Examples of international wargames focusing on space capabilities

While wargaming is recognised as providing value to European space operations in a military context, efforts for greater cohesion and integration are needed in the coming years. The EU SSSD openly calls for a closer connection between space, security, and defence at the EU level to bolster the EU Space Threat Response Architecture and to develop **regular exercises involving**

[9] EEAS, "Space: EU carries out Space Threat Response Architecture 2024 Exercise (STRA-X-24)", 13 Mar 2024. (Link)
[10] NATO Science and Technology Organisation, "NATO STO and HQ SACT conduct multi-domain space deterrence wargame", 16 May, 2024, (Link).
[11] Ministère des armées, "AsterX 2025: defending space together", 17 March 2025. (Link)
[12] NATO, "French Space exercise AsterX builds on realistic scenario and integration", 9 March 2023, (Link).
[13] UK Ministry of Defence, "Defence Space Strategy: Operationalising the Space Domain" (Link)

the space domain.[14] The Strategy also argues for organising EU-NATO joint exercises, which would include space as a component. Moreover, the Commission's 2022 Strategic Compass explicitly addressed that the Union should conduct exercises to test the resilience of space assets, most notably highlighting the validation of Galileo's threat response mechanisms.[15] Despite these statements, there remains a **lack of clarity regarding what constitutes (military) exercises in the space domain** in the eyes of the EU.

Europe needs to step up when it comes to coordinating military space activities. Wargaming can provide a key solution to enhance cohesion between states. The 2024 editions of AsterX and STRA-X-24 were held nearly simultaneously in March 2024 (4-15 March and 4-13 March, respectively) and show the growing momentum and need for exercising. Whether the full benefit of both could be leveraged remains unclear, as little information on the latter has been made public. If not coordinated, it creates a concern, as not all countries on the continent maintain relevant military space capabilities or sufficient budgets for conducting wargames involving space assets. In this sense, **pan-European wargames could be an avenue to leverage all capabilities and provide smaller states with the opportunity to become better acquainted with the rising stakes and contemporary strategies** that are increasingly critical in the case of conflict.

Pan-European wargaming practices could enable shaping a European doctrine for military space operations and facilitate progress towards a **regional approach to addressing threats and responses in space**. Common agreement on red lines and establishing protocols for engagement are essential due to the multitude of states involved and the **interconnected and interdependent civil and military infrastructure** of states, the EU, ESA and EUMETSAT.

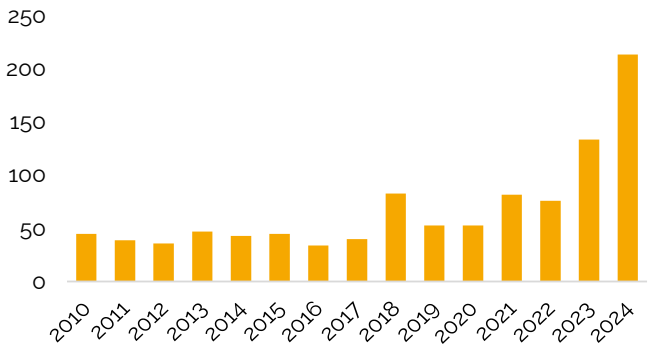


Figure 4: Number of military satellites launched since 2010

As the growth in military satellite launches indicates the rising importance states place on space for defence activities, delineating clear red lines and robust response mechanisms would bolster Europe's position as a global space actor. AsterX, with its ever-growing participation and the inclusion of key allies such as South Korea, Japan and Canada, is a welcome step towards a common approach on military operations in space among European and like-minded partners. **Broadening participation to governmental and industry players from other nations** can further strengthen collective decision-making and build awareness of shared needs. This, in turn, can help unlock greater political support and funding for federated and joint military space capabilities in Europe, including in partnership with key current and future allies.

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[14] EU Commission, 2023, "Joint Communication to the European Parliament and the Council – European Union Space Strategy for Security and Defence", JOIN(2023)9 (Link).
[15] European External Action Service, 2022, "A Strategic Compass for Security and Defence" (Link).